Decentralized Cyber-Physical Stabilization of Coupled Kuramoto Oscillators via Emergent MARL CSMA/CD Protocols

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Abstract

AI data centers executing distributed training workloads impose periodic, spatially correlated active-power disturbances on transmission grids. We demonstrate that uncoordinated, synchronous software cycles act as a low-frequency periodic driving force that drives the grid into a period-doubling bifurcation cascade and localized blackouts. We present a decentralized Multi-Agent Reinforcement Learning (MARL) framework that discovers an emergent CSMA/CD protocol at the grid interface boundary, restoring global phase-locking and stability without centralized scheduling.

1 Introduction and Problem Formulation

Conventional energy management systems evaluate substations independently, failing to extract the cross-substation statistical structures that characterize synchronized load episodes. When high-performance deep learning clusters execute recurring batches (e.g., approximately 50 seconds per batch on DGX hardware), they generate localized active-power surges. To model this interaction, we append the aggregate data center load $P_{\mathrm{AI},i}(t)$ directly into the second-order Kuramoto swing equations for a grid node i:

$$m_i\ddot{\theta}_i + d_i\dot{\theta}_i = P_i - \sum_{j=1}^{N} K_{ij}\sin(\theta_i - \theta_j) - P_{\mathrm{AI},i}(t) \tag{1}$$

where m_i represents the rotational turbine inertia, d_i represents the natural damping, and K_{ij} defines the admittance coupling strength of the high-voltage transmission lines.

2 Multi-Facility Scaling and Normalized Sensing

As the architecture scales up to 10,000 training nodes distributed across multiple locations at a density of 50 training nodes per facility, static global threshold sensing breaks down. The MARL agents adapt by implementing a normalized carrier-sensing metric $\overline{L}_i(t)$ tracking local facility capacity perimeters:

$$\overline{L}_i(t) = \frac{\sum_{j\in\mathcal{N}_i} P_{\mathrm{AI},j}(t)}{\mathrm{Capacity}_{\mathrm{facility}}} \tag{2}$$

This multi-scale localization allows individual clusters to sense the aggregate load acceleration ($\frac{dL}{dt}$) and execute independent out-of-phase gating delays, preventing the formation of chaotic attractors.

3 The Core MARL Framework and Emergent CSMA/CD

The decentralized controllers evaluate their state-action parameters using a unified, non-linear quadratic collective reward function:

$$R_{\text{collective}}(t) = \sum_{i=1}^{N} \text{Compute_Progress}_i(t) - \beta \cdot \max\left(0,\, L(t) - L_{\text{threshold}}\right)^2 \tag{3}$$

where $\beta = 1000$. Sensing an imminent collision via the load derivative, the agent introduces a randomized temporal delay gate that shifts the periodic square waves out of phase. The grid's natural damping d_i and mechanical inertia m_i recover dominance, dampening the frequency deviations $\dot{\theta}_i(t)$ and pulling the oscillators back into global synchronization.

4 Asymptotic Local Frequency Deviations Data Matrix

The data tables below contrast the open-loop trajectory towards a chaotic attractor with the closed-loop MARL intervention which collapses the multi-frequency cloud back into a stable phase-locked peak:

4.1 Phase 1: Open-Loop Repetitive Forcing (No Control)

FORCING AMP	ASYMPTOTIC FREQUENCY PEAKS (Hz)	REGIME
0.00 GW	24.965, 24.966, 24.980...	[CHAOTIC ATTRACTOR]
2.14 GW	21.438, 21.509, 21.916...	[CHAOTIC ATTRACTOR]
4.29 GW	18.844, 18.871, 18.885...	[CHAOTIC ATTRACTOR]
6.43 GW	15.720, 15.776, 16.024...	[CHAOTIC ATTRACTOR]
8.57 GW	12.747, 12.878, 12.886...	[CHAOTIC ATTRACTOR]
10.71 GW	9.715, 9.774, 9.893...	[CHAOTIC ATTRACTOR]
12.86 GW	6.677, 6.684, 6.714...	[CHAOTIC ATTRACTOR]
15.00 GW	3.681, 3.698, 3.714...	[CHAOTIC ATTRACTOR]

4.2 Phase 2: Closed-Loop MARL Access Gating Intervention

FORCING AMP	ASYMPTOTIC FREQUENCY PEAKS (Hz)	REGIME
0.00 GW	24.965	[PHASE-LOCKED STATE]
2.14 GW	24.964	[PHASE-LOCKED STATE]
4.29 GW	24.971	[PHASE-LOCKED STATE]
6.43 GW	24.954	[PHASE-LOCKED STATE]
8.57 GW	24.965	[PHASE-LOCKED STATE]
10.71 GW	24.942	[PHASE-LOCKED STATE]
12.86 GW	24.948	[PHASE-LOCKED STATE]
15.00 GW	24.943	[PHASE-LOCKED STATE]

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